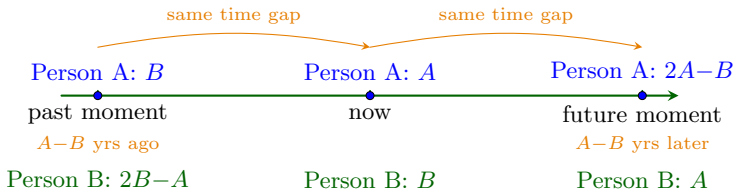
MCC Entrance Test, August 2026 — Day 1 — Solutions

Level T2

Test Information. Level T2 — 120 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Combinatorics (×2), Geometry, Number Theory.
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] Person A is currently five times as old as Person B was at the exact moment Person A was as old as Person B is now. By the time Person B grows old enough to reach Person A's current age, the sum of their ages at that time will be 72. How old is Person A right now?

Official solution. Let A be Person A's current age and B be Person B's current age. If the moment in question was t years ago, Person A's age back then was $A - t$; setting this equal to Person B's current age B gives $t = A - B$, so that moment was $A - B$ years ago, at which time Person B's age was $B - (A - B) = 2B - A$.



The first condition says

$$A = 5(2B - A) \implies A = 10B - 5A \implies 6A = 10B \implies B = \tfrac{3A}{5}. \tag{1}$$

Person B is B now and needs to reach age A, so Person B reaches Person A's current age $A - B$ years from now; at that point Person A's age will be $A + (A - B) = 2A - B$. The second condition says

$$A + (2A - B) = 72 \implies 3A - B = 72 \overset{(1)}{\implies} 3A - \tfrac{3A}{5} = 72 \implies A = \boxed{30}.$$

□

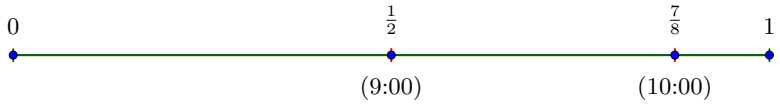
Problem 2 (10 pts).[2] A code is a sequence of four digits, each chosen from 1 through 9 (the digit 0 is not allowed). The first two digits, read together, must form a two-digit multiple of eleven; so must the last two digits. Finally, the sum of all four digits must itself be a multiple of eleven. How many codes satisfy all three conditions?

Official solution. A two-digit multiple of eleven with nonzero digits has the form $\overline{aa}$ (that is, $11, 22, \ldots, 99$), since these are exactly $11 \cdot 1, 11 \cdot 2, \ldots, 11 \cdot 9$. So the code has the form $\overline{a\,a\,b\,b}$ for digits $a, b \in \{1, \ldots, 9\}$.

The digit sum is $2a + 2b = 2(a + b)$. Since $\gcd(2, 11) = 1$, this is a multiple of 11 exactly when $a + b$ is a multiple of 11. As $a, b \in \{1, \ldots, 9\}$, the sum $a + b$ ranges from 2 to 18, so the only possibility is $a + b = 11$.

The pairs (a, b) with $a, b \in \{1, \ldots, 9\}$ and $a + b = 11$ are $(2, 9), (3, 8), (4, 7), (5, 6), (6, 5), (7, 4), (8, 3), (9, 2)$ – that is, a ranges over $2, 3, \ldots, 9$, giving $\boxed{8}$ codes. □

Problem 3 (15 pts).[3] A task proceeds at a constant rate. By 9:00 p.m., $\frac{1}{2}$ of the task was complete; by 10:00 p.m., $\frac{7}{8}$ of the task was complete.



(a) At what fractional rate (of the task per hour) is the task proceeding? [4 pts]

(b) At this same rate, when will the task be completely finished? [11 pts]

[1] Purple Comet! Math Meet, Middle School Division. [2] UK Junior Mathematical Challenge. [3] Gauss Contest, Centre for Education in Mathematics and Computing.

Official solution. **Part (a).** Between 9:00 and 10:00 (one hour), the fraction complete increased from $\frac{1}{2}$ to $\frac{7}{8}$, an increase of $\frac{7}{8} - \frac{1}{2} = \frac{3}{8}$. So the task proceeds at a rate of $\boxed{\frac{3}{8}}$ per hour.

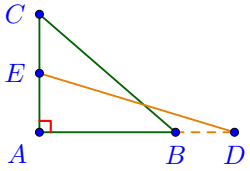
Part (b). After 10:00, the fraction remaining is $1 - \frac{7}{8} = \frac{1}{8}$. At a rate of $\frac{3}{8}$ per hour, the time needed is $\frac{1/8}{3/8} = \frac{1}{3}$ hour = 20 minutes. So the task is finished at $\boxed{10{:}20 \text{ p.m.}}$ □

Problem 4 (15 pts).

[4] In right triangle ABC with the right angle at A, leg AB has unit length. Point D lies on the extension of AB beyond B so that $BD = \frac{1}{2}AC$. Let E be the midpoint of leg AC. Given that the ratio of areas $[AED] : [ABC] = 2 : 3$,

(a) Let $AC = b$. Express $[ABC]$ and $[AED]$ in terms of b. [6 pts]

(b) Solve for b, then find the length of BD. [9 pts]



Official solution. **Part (a).** Since $\angle BAC = 90°$, $[ABC] = \frac{1}{2} \cdot AB \cdot AC = \frac{1}{2} \cdot 1 \cdot b = \frac{b}{2}$.
Since E lies on ray AC and D lies on ray AB, the angle $\angle EAD$ is the same right angle $\angle CAB = 90°$.

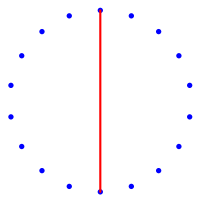
$$AE = \tfrac{b}{2},\ AD = AB + BD = 1 + \tfrac{b}{2} \implies [AED] = \tfrac{1}{2} \cdot AE \cdot AD = \tfrac{1}{2} \cdot \tfrac{b}{2}\left(1 + \tfrac{b}{2}\right) = \boxed{\frac{b}{4}\left(1 + \frac{b}{2}\right)}.$$

Part (b). Using part (a), and cancelling the common factor $\frac{b}{2}$ from numerator and denominator,

$$\frac{[AED]}{[ABC]} = \frac{\frac{b}{4}\left(1 + \frac{b}{2}\right)}{\frac{b}{2}} = \frac{1 + \frac{b}{2}}{2} = \frac{2 + b}{4} \implies \frac{2 + b}{4} = \frac{2}{3} \implies b = \frac{2}{3} \implies BD = \tfrac{1}{2}AC = \boxed{\tfrac{1}{3}}.$$

□

Problem 5 (25 pts).[5] Eighteen points are equally spaced around a circle. Some of the points are marked. A *marked rectangle* forms whenever four of the marked points sit at the four corners of a rectangle.



one diametrically-opposite pair

(a) Prove that the four corners of any marked rectangle form two diametrically-opposite pairs of points, and find how many such pairs exist among the 18 points. [12 pts]

(b) What is the largest number of points that can be marked while completing *at most one* full diametrically-opposite pair? [4 pts]

(c) Hence determine the minimum number of points that must be marked to *guarantee* that a marked rectangle has formed. [9 pts]

Official solution. **Part (a).** All 18 points lie on a circle, so any rectangle with all four corners among them is inscribed in that circle. A rectangle's diagonals bisect each other and have equal length, so their common midpoint is equidistant from all four (concyclic) corners – and the only point equidistant from points on a circle is the circle's own center. So each diagonal passes through the center with its two endpoints equidistant from it, i.e. each diagonal is a diameter. A rectangle's two diagonals connect opposite corners, so the four corners split into exactly two diametrically-opposite pairs.

Since the 18 points are equally spaced, each point has exactly one diametrically-opposite partner (the point 9 positions away), giving $18/2 = \boxed{9}$ such pairs.

[4] KöMaL (Középiskolai Matematikai és Fizikai Lapok). [5] Australian Mathematics Competition.

Part (b). To keep at most one pair fully marked, one can mark one point from *each* of the 9 pairs (9 points, 0 complete pairs), then mark the partner of just one already-marked point (completing exactly 1 pair). That's $9 + 1 = \boxed{10}$ points with only one complete pair – not yet a rectangle.

Part (c). By part (b), 10 points can be marked with only one complete diametrically-opposite pair. Every one of the 9 pairs already has at least one point marked, so the very next (11th) point must be the missing partner of some *other* pair, completing a second full pair. Two complete pairs give the four corners of a marked rectangle (by part (a)).

So $\boxed{11}$ points must be marked to guarantee a marked rectangle forms. □

Problem 6 (25 pts).[6] A six-digit number $\overline{2ABCDE}$, where A, B, C, D, E are unknown digits, has the following property. When this number is tripled, it becomes exactly three times as large – and its digits cycle: the leading 2 moves to the very end, giving the new number $\overline{ABCDE2}$.

(a) Let $N = \overline{ABCDE}$ denote the five-digit number formed by digits A through E (some may be leading zeros within N). What equation must N satisfy? [9 pts]

(b) Without dividing, find the last (units) digit of N. [6 pts]

(c) Hence find N, and the sum of the digits of the number $\overline{2ABCDE}$. [10 pts]

Official solution. **Part (a).** The original number is $\overline{2ABCDE} = 200000 + N$. Tripling it gives the new number $\overline{ABCDE2} = 10N + 2$ (shift N one place left and append the digit 2). The tripling condition says

$$3(200000 + N) = 10N + 2 \implies 600000 + 3N = 10N + 2 \implies 7N = \boxed{599998}.$$

Part (b). Reducing $7N = 599998$ modulo 10 gives $7N \equiv 8 \pmod{10}$. Testing each possible last digit d of N, the values of $7d \bmod 10$ for $d = 0, 1, \ldots, 9$ are $0, 7, 4, 1, 8, 5, 2, 9, 6, 3$ – so $7d \equiv 8 \pmod{10}$ exactly when $d = \boxed{4}$.

Part (c). From part (a), $N = \dfrac{599998}{7} = 85714$ (note its last digit, 4, matches part (b)). The original number is $200000 + 85714 = 285714$. Its digits are $2, 8, 5, 7, 1, 4$, so their sum is

$$2 + 8 + 5 + 7 + 1 + 4 = \boxed{27}.$$

(Check: $285714 \times 3 = 857142 = \overline{ABCDE2}$ with $N = 85714$, confirming the cycle – and since 857142 is just a rearrangement of the same six digits, it has the same digit sum, 27.) □

[6] Kangaroo Math Competition (Kangourou Sans Frontières).